LJ University
University with a Difference

LJ INNOVATION VILLAGE

MEET THE TEAM

PRIYANI RATHOD
SEM - 1 : BE - AIDS

PUSHTI TRAMBADIYA
SEM - 1 : BE - IT

RAVI TRIVEDI
SEM - 1 : BE-IT

HET VEDANI
SEM - 1 : BE - IT

SURYANSH CHANDAK
SEM - 1 : BE -AIDS

LET'S CONNECT

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LJ INSTITUTE OF ENGINEERING AND TECHNOLOGY

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Feel to Read : A Braille companion

"Empowering Learning Through Touch."

The project is an Arduino-based Braille learning system that converts user input into tactile Braille patterns using solenoids. It applies electromagnetism and embedded system concepts to enable quiet, low-cost, and independent learning for visually impaired users, making it suitable for inclusive educational environments.

01

WHY ARE WE DOING THIS PROJECT? (PROJECT RATIONALE) .

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Visually impaired students face majo rdifficulty accessing printed learning materials. Braille books are expensive, bulky, and slow to produce. Our project provides a low-cost, real-time solution that converts normal text into tactile Braille, enabling independent learning and improved accessibility.

02

WHICH PROBLEM DOES IT ADDRESS?

Manyvisually impaired learners cannot read printed text without assistance. Existing Braille materials require long preparation time and are not always available. This creates dependency on others and limits educational opportunities. Our system helps by instantly converting digital text into physical Braille dots.

03

WHAT ARE THE ALTERNATE SOLUTIONS OUT THERE?

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Current alternatives include printed Braille textbooks, refresha Braille displays, and audio-based screen readers. While effect these solutions are often costly, complex, and not affordable fo many institutions or individuals. Some systems are bulky and require specialized infrastructure.

04

WHAT MAKES OUR PROJECT STAND OUT FROM THE REST? THIS IS YOUR UNIQUENESS OR INNOVATION.

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Our project is unique because it is affordable, simple, and suitable for education. It generates Braille in real time using easily available components independent dot control, demonstrates embedded systems and electromagnetism, and can be deployed in schools and colleges at much lower cost.